Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

1 / 1 point

1. Which Google Cloud service is recommended as the primary, cost-effective object storage for a data lakehouse, capable of handling structured, semi-structured, and unstructured data?

- (selected) Cloud Storage
- Apache Iceberg
- BigQuery
- AlloyDB

Correct
Correct! Google Cloud Storage is the primary object storage service used to build the foundation of a data lake. It's designed to cost-effectively store enormous volumes of raw data in any format.

1 / 1 point

2. For an online retail business like Cymbal, which of the following use cases is best suited for AlloyDB?

- Storing raw, unstructured log files generated from website clicks.
- Performing complex analytical queries on historical marketing campaign performance.
- (selected) Managing real-time customer order processing and inventory updates.
- Storing large volumes of historical sales data for quarterly trend analysis.

Correct
Correct! AlloyDB is a high-performance, PostgreSQL-compatible database built for high-throughput transactional workloads (OLTP) like managing orders, customer data, and inventory in real-time.

1 / 1 point

3. What is the primary role of Apache Iceberg in a data lakehouse architecture built on Google Cloud?

- (selected) It adds a metadata layer to files in Cloud Storage, enabling features like schema evolution, time travel, and efficient querying.
- It is the primary service used to store raw, unprocessed data files of any format or size.
- It acts as the central query engine for running high-speed SQL analytics.
- It is a fully managed, PostgreSQL-compatible database for operational workloads.

Correct
Correct! Apache Iceberg is an open table format that brings warehouse-like capabilities (ACID transactions, schema management, time travel) to files stored in a data lake on Cloud Storage, making them more reliable and manageable.

1 / 1 point

4. How does BigQuery enable a unified analytics platform in the Cymbal company's data lakehouse?

- By requiring all data to be moved from Cloud Storage and AlloyDB into its native storage.
- By converting all unstructured and semi-structured data into a single proprietary format before analysis.
- By exclusively managing high-transaction operational workloads with low latency.
- (selected) By using federated queries to analyze data directly in external sources like AlloyDB and in Iceberg tables on Cloud Storage without moving the data.

Correct
Correct! BigQuery's ability to run federated queries allows it to act as a single pane of glass for analytics. It can query data where it lives—transactional data in AlloyDB, raw data in Cloud Storage managed by Iceberg, and its own native storage—creating a unified platform without costly data duplication.